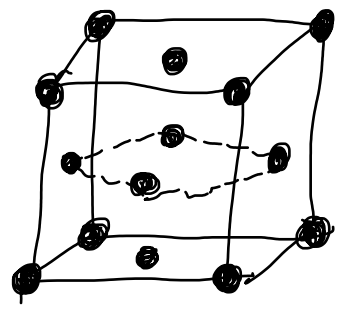
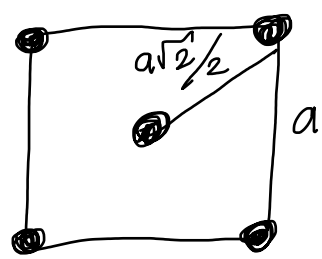
PRACTICE TEST SOLUTIONS

1. **(7 points)** Consider a face-centered cubic (fcc) lattice:

   (a) (2 points) Draw a 3D primitive unit cell structure



   (b) (2 points) Sketch the placement of atoms on a (100) plane. Express distances between atoms on the plane in terms of lattice constant (a).



   (c) (1 point) How many atoms are there per primitive unit cell?

   4

   (d) (1 point) How many nearest neighbor atoms are there for each atom?

   12

   (e) (1 point) Assume a lattice constant of 1nm. Determine the number of atoms/cm^3

   $4 \text{ atoms}/\text{nm}^3 = \boxed{4 \times 10^{21} \text{ atoms}/\text{cm}^3}$

2. **(5 points)** A silicon wafer is doped with $10^{15}\ cm^{-3}$ boron atoms.

(a) (1 point) Is the wafer n- or p-type?

p-type

(b) (2 points) Calculate the electron and hole densities at 0 K.

$n = 0$

$p = 0$

(c) (2 points) Calculate the electron and hole densities at room temperature.

$p = 10^{15}/cm^3$

$n = n_i^2/10^{15} = \frac{10^{20}}{10^{15}} = 10^5/cm^3$

$\boxed{p = 10^{15}/cm^3 \quad n = 10^5/cm^3}$

3. **(4 points)** What is the dominant current mechanism in

(a) (2 points) forward biased p-n junction

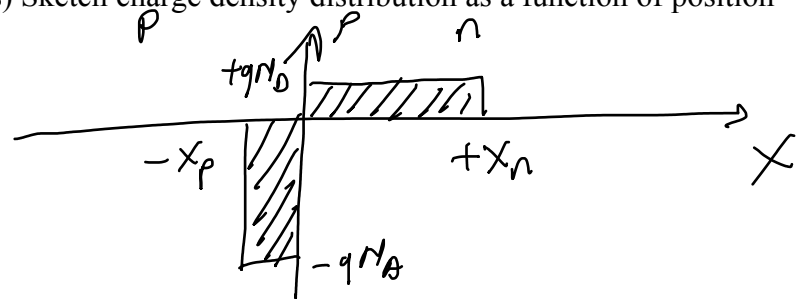
majority carrier diffusion

(b) (2 points) reverse biased p-n junction

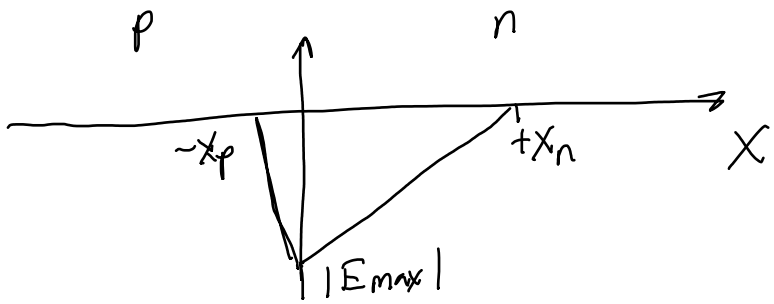
minority carrier drift

4. **(13 points)** Consider a p-n step junction under equilibrium with **p-side more heavily doped**. Label all necessary parameters on your sketches.

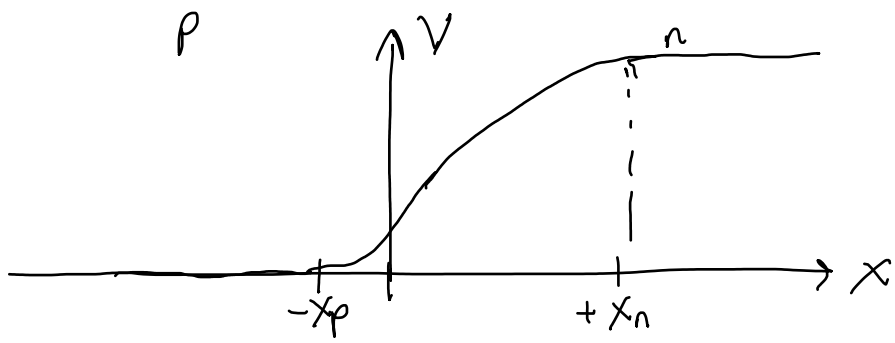
(c) (3 points) Sketch charge density distribution as a function of position



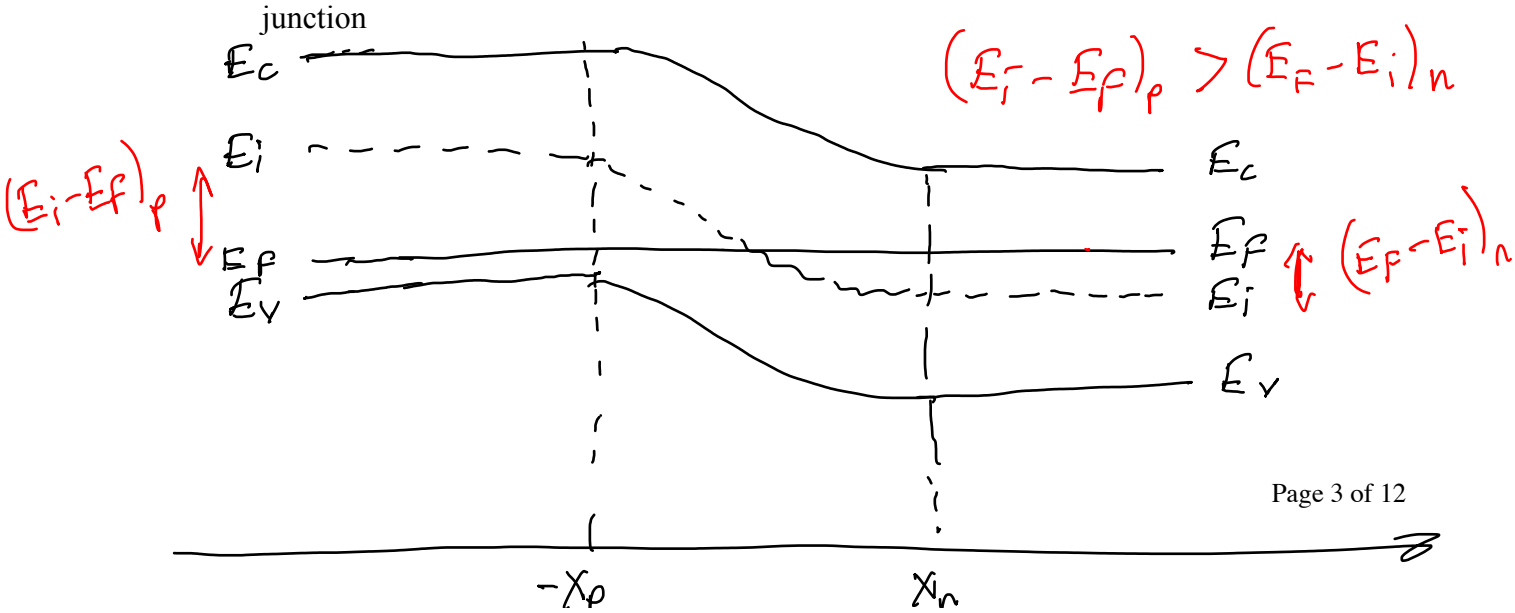
(d) (3 points) Sketch the electric field as a function of position



(e) (3 points) Sketch the electrostatic potential as a function of position



(f) (4 points) Sketch the energy band diagram labeling E_C, E_V, E_i and E_F on both sides of the junction



5. **(6 points)** Assume that a p-type and an n-type silicon semiconductor are nondegenerately doped with the same doping concentration. Between two samples, the p-type has higher resistivity than the n-type at room temperature.

(a) (3 points) How does electron and hole mobilities compare? Explain.

$$\rho_p = \frac{1}{q\mu_p N_A} \qquad \rho_p > \rho_n \Rightarrow \frac{1}{q\mu_p N} > \frac{1}{q\mu_n N} \Rightarrow \boxed{\mu_n > \mu_p}$$

$$\rho_n = \frac{1}{q\mu_n N_D}$$

(b) (3 points) How does electron and hole diffusion coefficients compare? Explain.

Because of Einstein relations

$$D_n = \frac{kT}{q}\mu_n \quad , \quad D_p = \frac{kT}{q}\mu_p \Rightarrow \boxed{D_n > D_p}$$

6. **(5 points)** Why are direct bandgap materials more efficient in absorbing light compared to indirect bandgap materials? Explain

In direct bandgap materials, only energy needs to be conserved. Therefore, band to band generation can be achieved with a photon.

In indirect bandgap semiconductors, both momentum and energy needs to be conserved. For band-to-band generation both photon and phonons should be involved in the process making the process less efficient.

7. **(4 points)** Which capacitance dominates the small signal behavior of a reverse biased p-n junction and what is the origin of this capacitance?

Junction or depletion layer capacitance (C_j)

C_j arises due to majority carrier oscillations at the edges of the depletion region.

8. **(6 points)** Consider a p-n junction

(a) (3 points) What is the physical mechanism leading to an avalanche breakdown in a p-n junction? Explain the mechanism.

Impact ionization: Carriers accelerated under E-field impact with lattice atoms to create new electron hole pairs which gain energy to create more electron-hole pairs

(b) (3 points) How does breakdown voltage in a reverse-biased p-n junction change with doping concentration of the lightly doped side? Explain why?

$N_B \searrow$ $V_{br} \nearrow$ because wider depletion region due to lightly doped p-n junction requires greater voltage to achieve the required E-field for breakdown.

Problem (50 points): A silicon step junction at T = 300 K has doping concentrations of $N_A = 5x10^{16}\ cm^{-3}$ and $N_D = 5x10^{15}\ cm^{-3}$. The cross-sectional area of the junction is 100 μm^2.
Show all steps in your solution

(a) (5 points) Calculate the Fermi energy on each side of the junction (in the quasi-neutral regimes) with respect to the intrinsic Fermi level, i.e. E_F - E_i.

(b) (5 points) Calculate the resistivity of both n-type and p-type quasi-neutral regions.

(c) (5 points) Calculate the built-in potential V_{bi}.

(d) (5 points) Calculate the width of the depletion region under zero bias. What percentage of the depletion region is in the n-side of the junction?

(e) (5 points) By what factor does the depletion region width increase if you reverse bias the junction by at V_A = -25 V?

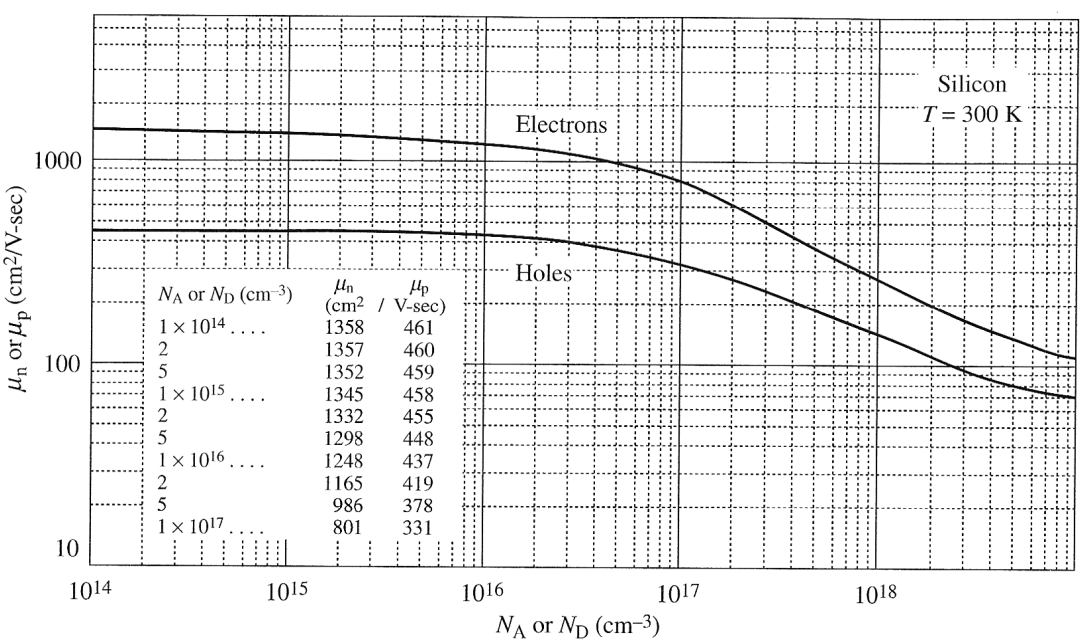
(f) (5 points) Calculate junction capacitance of p-n junction reverse biased at V_A = -25 V.

(g) (10 points) Now assume that the p-n junction is forward biased with V_A = 0.5 V. Sketch, in log scale, the steady-state majority and minority carrier concentrations on both sides of the depletion region as a function of the position. Indicate the equilibrium level of minority carrier concentrations in your sketch.

(h) (10 points) Calculate the current density flowing through the p-n junction in case of a forward-bias voltage of V_A = 0.5 V. Assume a minority carrier lifetime for holes and electrons of $\tau_n = \tau_p$ = 1 μs.

You might need:

$q = 1.6x10^{-19}$ C, $K_s = 11.8$, $\varepsilon_0 = 8.85x10^{-14}$ F/cm, $kT = 0.026$ eV at T = 300K
$E_g = 1.12$ eV, $n_i = 10^{10} cm^{-3}$ at T = 300K, use carrier mobilities from the graph below.



(Use for calculations)

(a) on the n-side $n \approx N_D = 5 \times 10^{15} / cm^3$

$$n = n_i e^{(E_F - E_i)/kT} \Rightarrow (E_F - E_i)_{n\text{-side}} = kT \ln\left(\frac{N_D}{n_i}\right)$$

$$\boxed{\cong 0.341 \, eV}$$

on the p-side $p \approx N_A = 5 \times 10^{16} / cm^3$

$$p = n_i e^{(E_i - E_F)/kT} \Rightarrow (E_i - E_F)_{p\text{-side}} = kT \ln\left(\frac{N_A}{n_i}\right)$$

$$\boxed{= 0.401 \, eV}$$

(b) On the n-side $\rho = \dfrac{1}{q \mu_n N_D} = \boxed{0.963 \ \Omega cm}$

On the p-side $\rho = \dfrac{1}{q \mu_p N_A} = \boxed{0.331 \ \Omega cm}$

(Use for calculations)

(c)

$$V_{bi} = \frac{1}{q}\left(E_p - E_i\right)_{n\text{-side}} + \frac{1}{q}\left(E_i - E_p\right)_{p\text{-side}}$$

$$= 0.341 + 0.401 = \boxed{0.742\ V}$$

(d)

$$W = \left[\frac{2K_s\varepsilon_0}{q} \cdot \frac{N_A + N_D}{N_A N_D}\left(V_{bi} - V_A\right)\right]^{1/2}$$

For $V_A = 0$ $\boxed{W = 0.462\ \mu m}$

$$x_n N_D = x_p N_A \Rightarrow x_n = \frac{N_A}{N_D} x_p$$

$$x_n = 10\, x_p \Rightarrow \frac{x_n}{W} = \frac{10 x_p}{10 x_p + x_p}$$

$$= \frac{10}{11} \approx \boxed{91\ \%}$$

(Use for calculations)

(e) For $V_A = -25\,V$

$$\frac{W(V_A = -25V)}{W(V_A = 0V)} = \left[\frac{V_{bi} + 25}{V_{bi}}\right]^{1/2}$$

$$\cong \boxed{5.89 \text{ times}}$$

(f)

$$C_j = \frac{K_s \varepsilon_0 A}{W(V_A = -25V)}$$

$$\cong 3.84 \times 10^{-15} F = \boxed{3.84\ fF}$$

(Use for calculations)

(g) $n_n = N_D = 5\times10^{15}/cm^3$

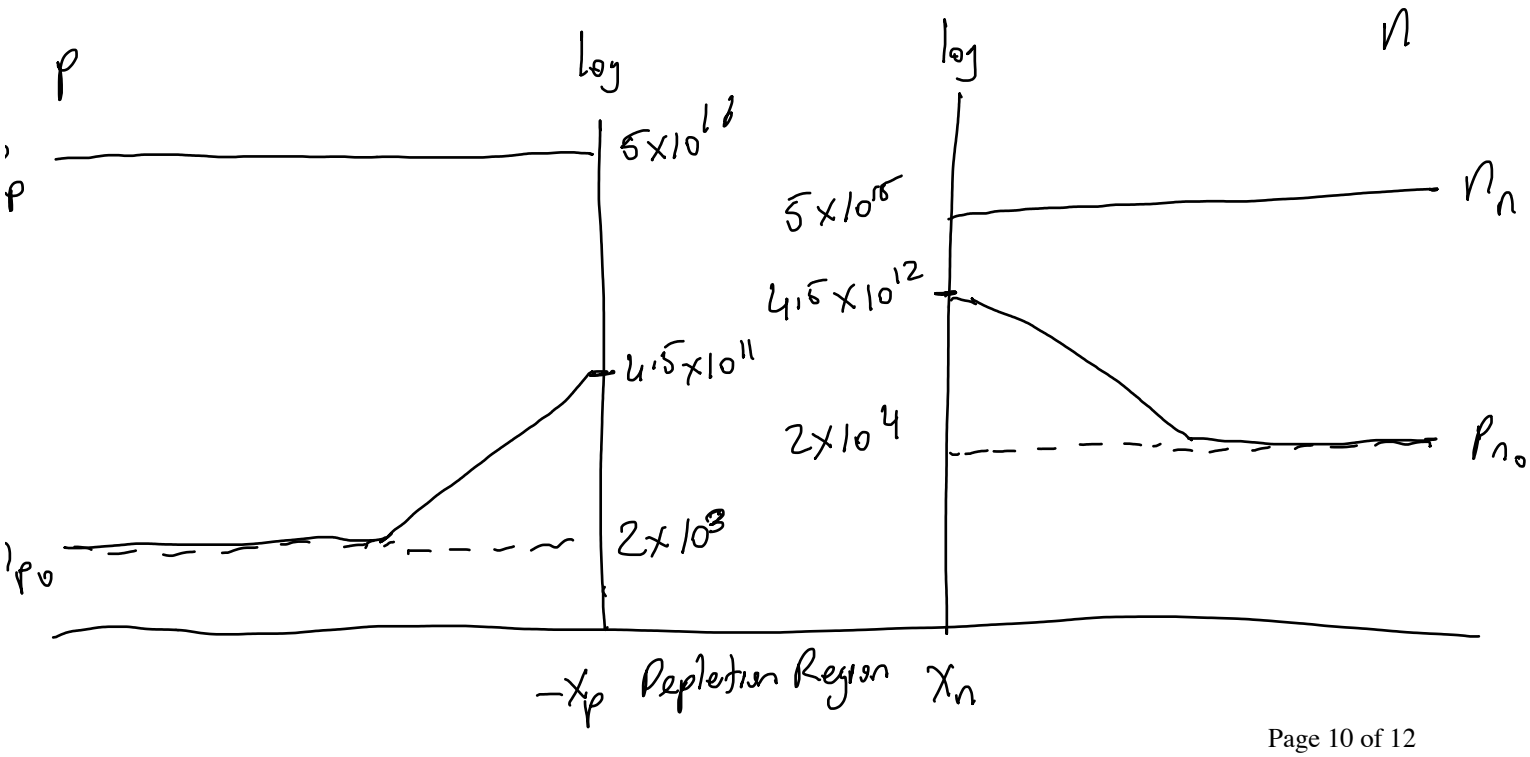
$$\Rightarrow p_{n_0} = n_i^2/N_D = 2\times10^4/cm^3$$

$$p_p = N_A = 5\times10^{16}/cm^3$$

$$\Rightarrow n_{p_0} = n_i^2/N_A = 2\times10^3/cm^3$$

$$n_p(-x_p) = n_{p_0}\, e^{qV_A/kT} \approx 4.5\times10^{11}/cm^3$$

$$p_n(x_n) = p_{n_0}\, e^{qV_A/kT} \approx 4.5\times10^{12}/cm^3$$



(Use for calculations)

(h)

$$J = \underbrace{\left[\frac{q D_n n_{p_0}}{L_n} + \frac{q D_p p_{n_0}}{L_p} \right]}_{J_s} \left[e^{qV_A/kT} - 1 \right]$$

$$D_n = \frac{kT}{q} \mu_n \cong 25.64 \ cm^2/s$$

$\uparrow$ 986 $cm^2/V.s.$

$$D_p = \frac{kT}{q} \mu_p \cong 11.65 \ cm^2/s$$

$\uparrow$ 448 $cm^2/V.s.$

$$L_n = \sqrt{D_n \tau_n} \approx 50.6 \ \mu m$$

$$L_p = \sqrt{D_p \tau_p} \approx 34.1 \ \mu m$$

$$J_s = 1.255 \times 10^{-11} \ A/cm^2$$

For $V_A = 0.5V$ $\boxed{J = 2.82 \ mA/cm^2}$

(Use for calculations)